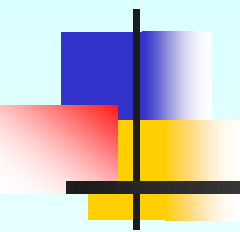
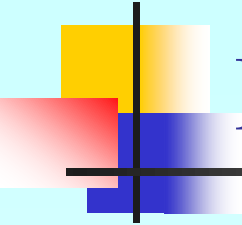


Lecture 3



Spatial Analysis with Vector Data

矢量数据的空间分析



Main Contents

1. **Buffer Analysis**
2. **Overlay Analysis**
3. **Network Analysis**
4. *Shortest route Analysis*

信 息 栏 学 年 级 姓 名

线

5. 给出下列图层及其属性表：灰色细边线的面 Pg1，其属性如表 1；无色粗边线的面 Pg2，其属性如表 2。写出叠置分析结果（Union，all 选项）的属性表（10 分）。

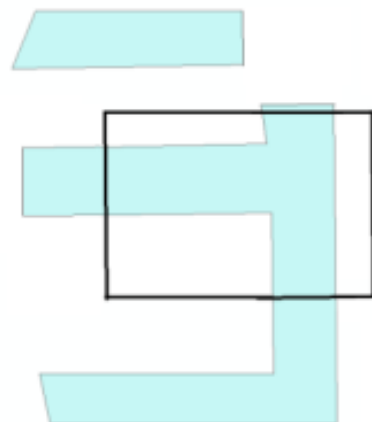
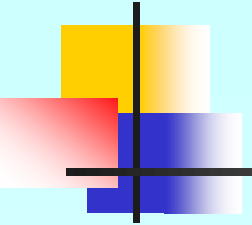


表 1

FID	Shape	Id
0	Polygon	m
1	Polygon	n

表 2

FID	Shape	Name
0	Polygon	A



1. Buffer Analysis

■ Basic Conception 基于点集拓扑学 Point Set Topology

For a given object A, the buffer area can be defined as follows:

$$P = \{ x \mid d(x, A) \leq R \}$$

d generally refers to Euler distance, it can also refer to other kind of distance. **R** is the radius of neighborhood or the condition of buffer created.

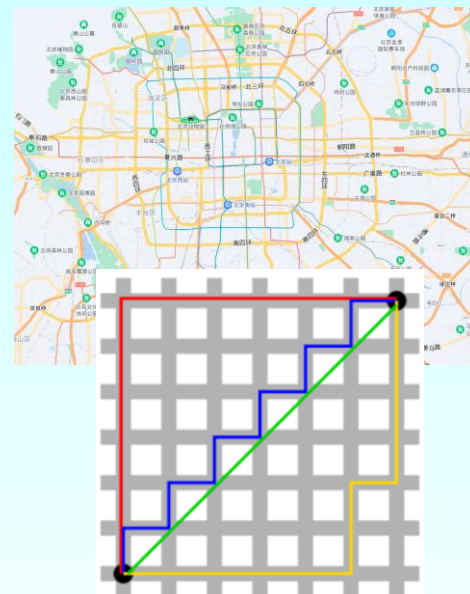
D: different definition of distance (数学严密性的优势, 概念对创新的重要性)

■ **R: different kind of radius**

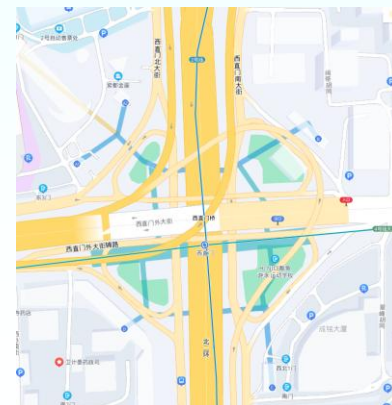
■ **A: Point, Polyline, Polygon, Volume, Temporal GIS** (时空对象)

About different kinds of distance

- Euclidean distance
- **Manhattan distance** (开放小区后城市街区可达性分析)
- Canberra distance 堪培拉~, 向量相似度
- Hausdorff distance 豪斯多夫~, 点集之间的距离
- Chebyshev distance 切比雪夫~, 国际象棋棋盘距离, 向量空间距离
- Kullback–Leibler distance KL散度, 概率分布之间的距离
- Levenshtein distance (Edit Distance)字典距离, 字符串之间的相似度
- Mahalanobis distance 马氏~, 数据点之间相关性
- Hamming distance 汉明~, 等长字符串之间的差异, 编码理论, DNA序列分析
- Lee distance 李氏~, 字符串之间的差异, 编码理论



曼哈顿距离的等价长度



1. Buffer Analysis

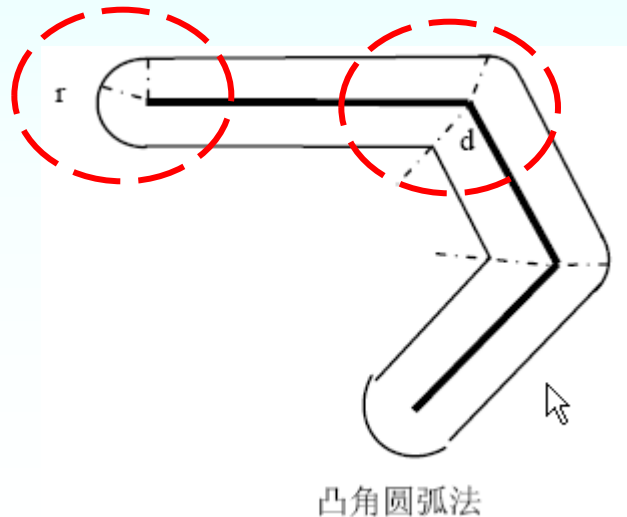
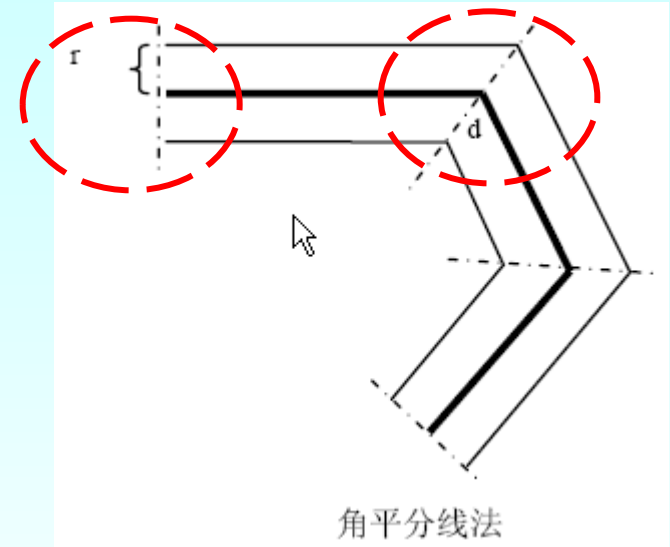
■ Essential problems

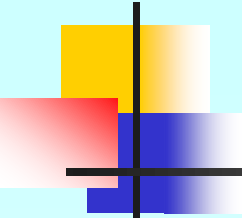
■ Method: exception of “ $\leq R$ ”

(奇异点)

■ 几何不连续

■ 属性突变 (如宽度)





1. Buffer Analysis

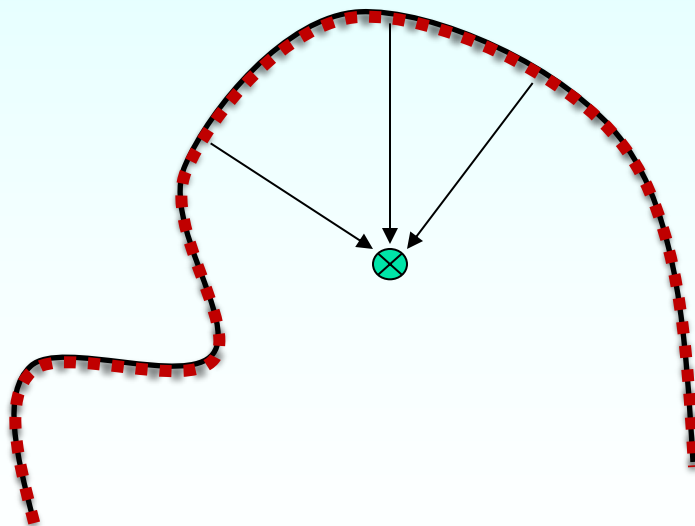
■ Essential problems

■ **Application:** (列举缓冲区分析的应用案例, 每个2分)

■ **Diffusion** 扩散分析

■ **Query**

■ **Planning**



1. Buffer Analysis

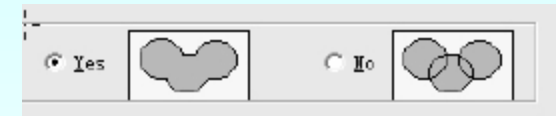
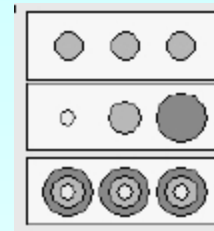
■ Tips of Buffering

■ Three kind of buffer

■ Merge or not

■ Inside / outside buffer of Polygon feature)

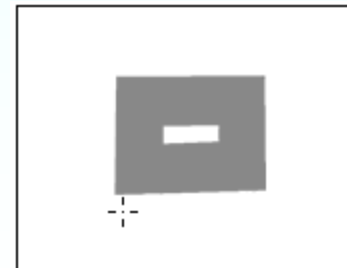
■ 对各种情形分别列举应用案例（如水域危险区划分）



inside and outside



only outside



only inside



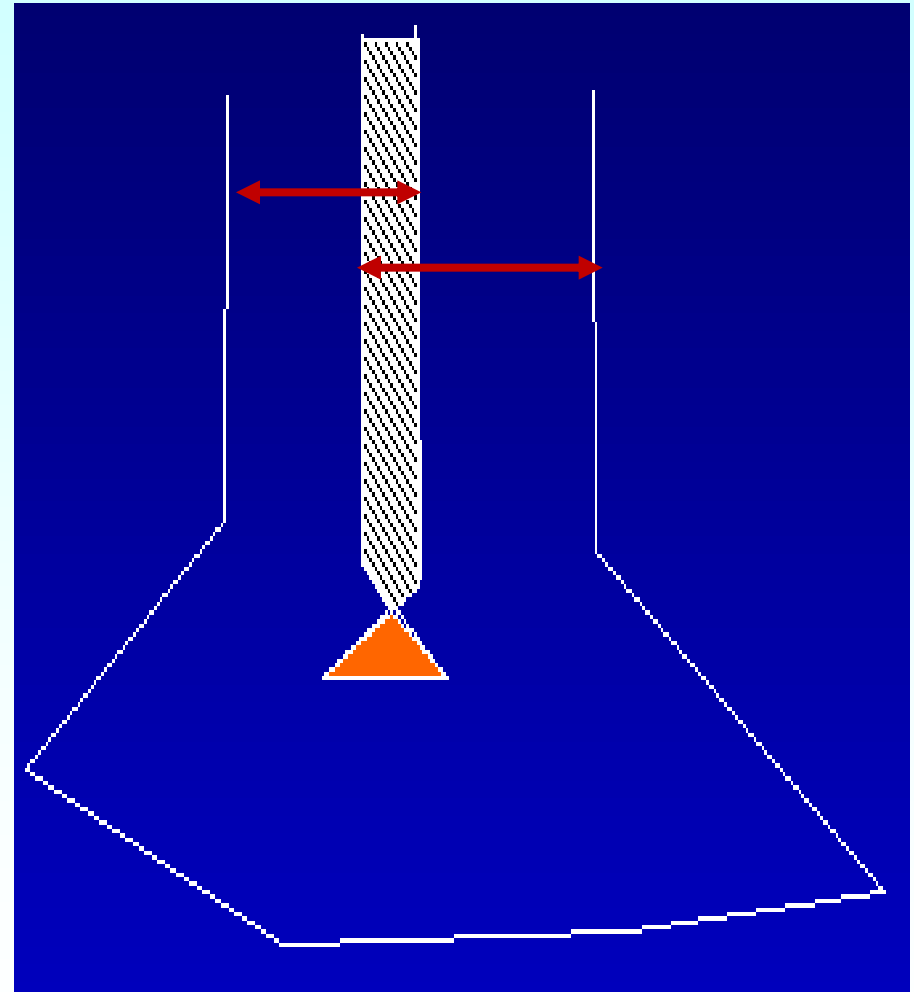
outside and include inside

1. Buffer Analysis

■ Bugs in buffering

- Topological error caused by buffering

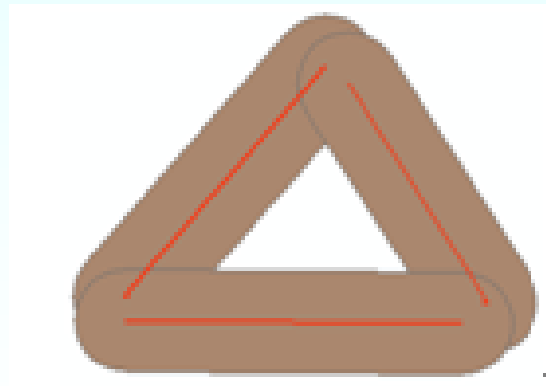
A cavum or self-crossing polygon will appear when the space between two polyline is too wide or too small for buffer area.



1. Buffer Analysis

■ Buffering with ArcGIS

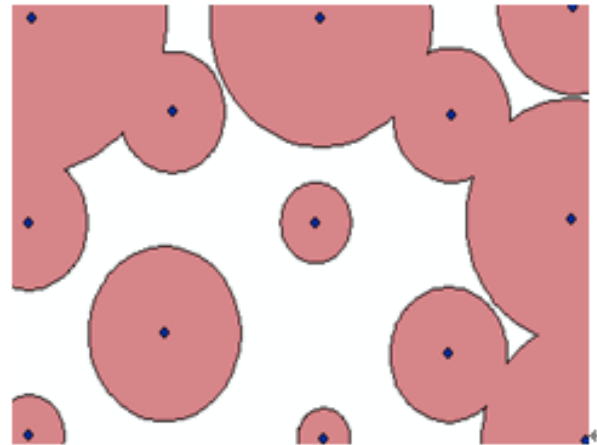
- **Application?** 缓冲区分析用于拓扑分析、数据处理。如点群的外包络线，膨胀腐蚀等。



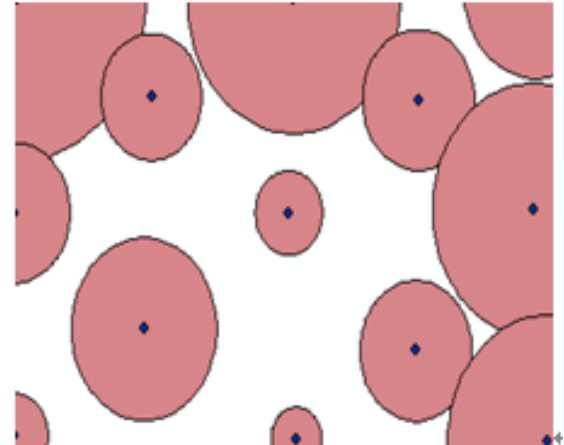
1. Buffer Analysis



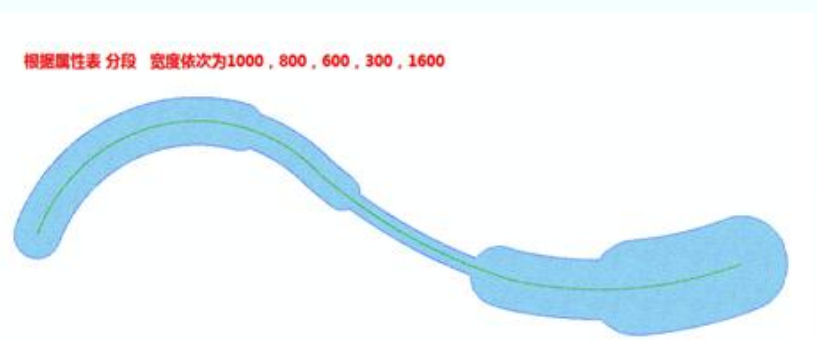
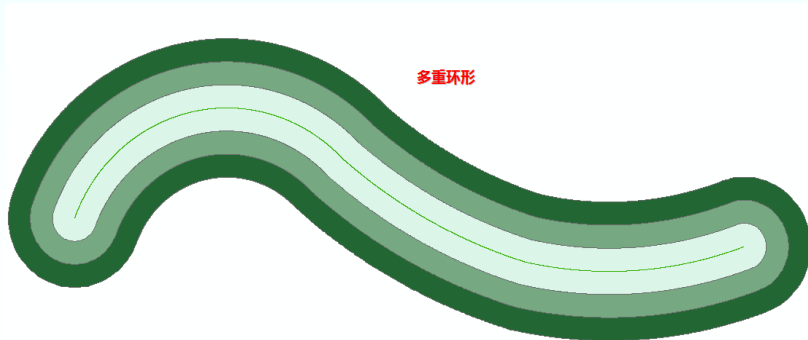
Graded Buffer

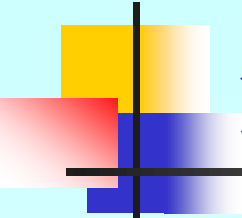


Dissolve barriers



Not dissolve barriers





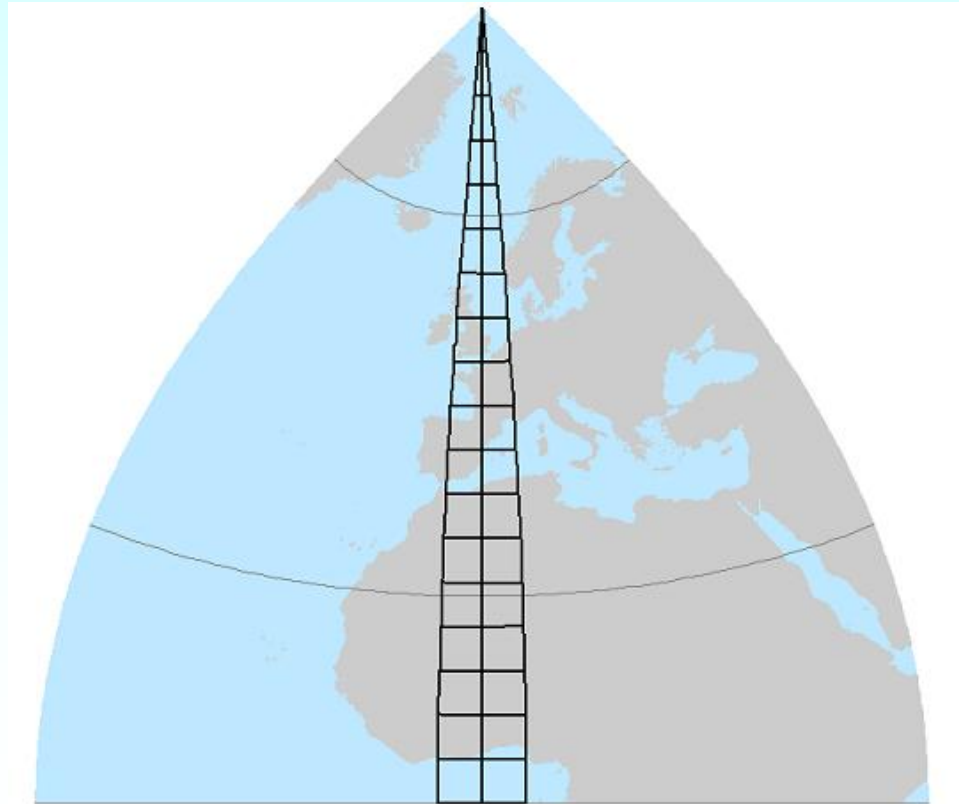
1. Buffer Analysis

问题1：缓冲区分析的结果，输出为面图层后，在不同的缓冲模型下，面对象的个数与缓冲对象的个数之间是何关系？

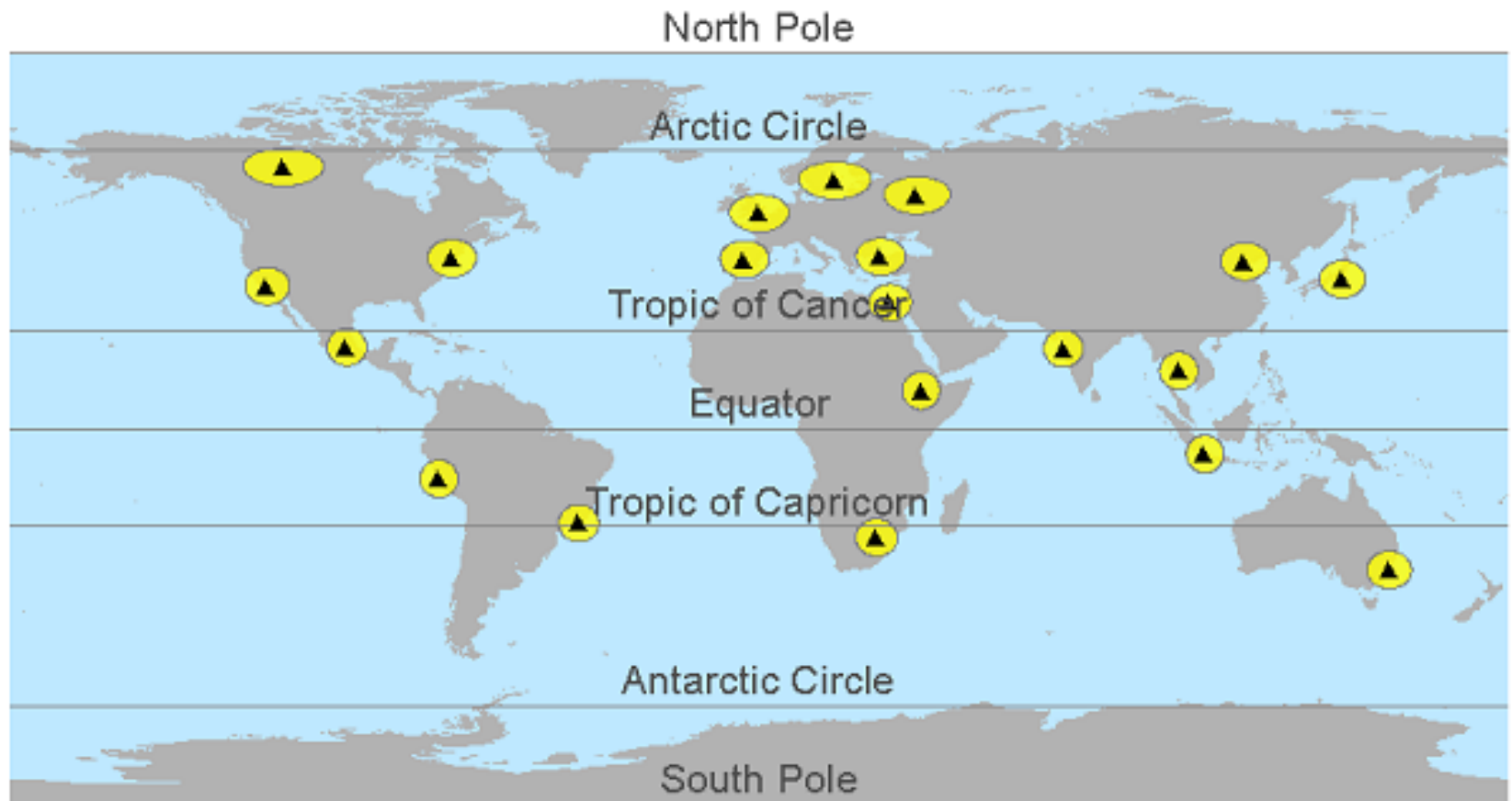
1. Buffer Analysis

问题2：数据的空间参考如何影响缓冲区分析的结果？

The graphic shows how the squares described by one degree of longitude and one degree of latitude vary in shape and size as you move further and further from the equator. (以度为距离单位)

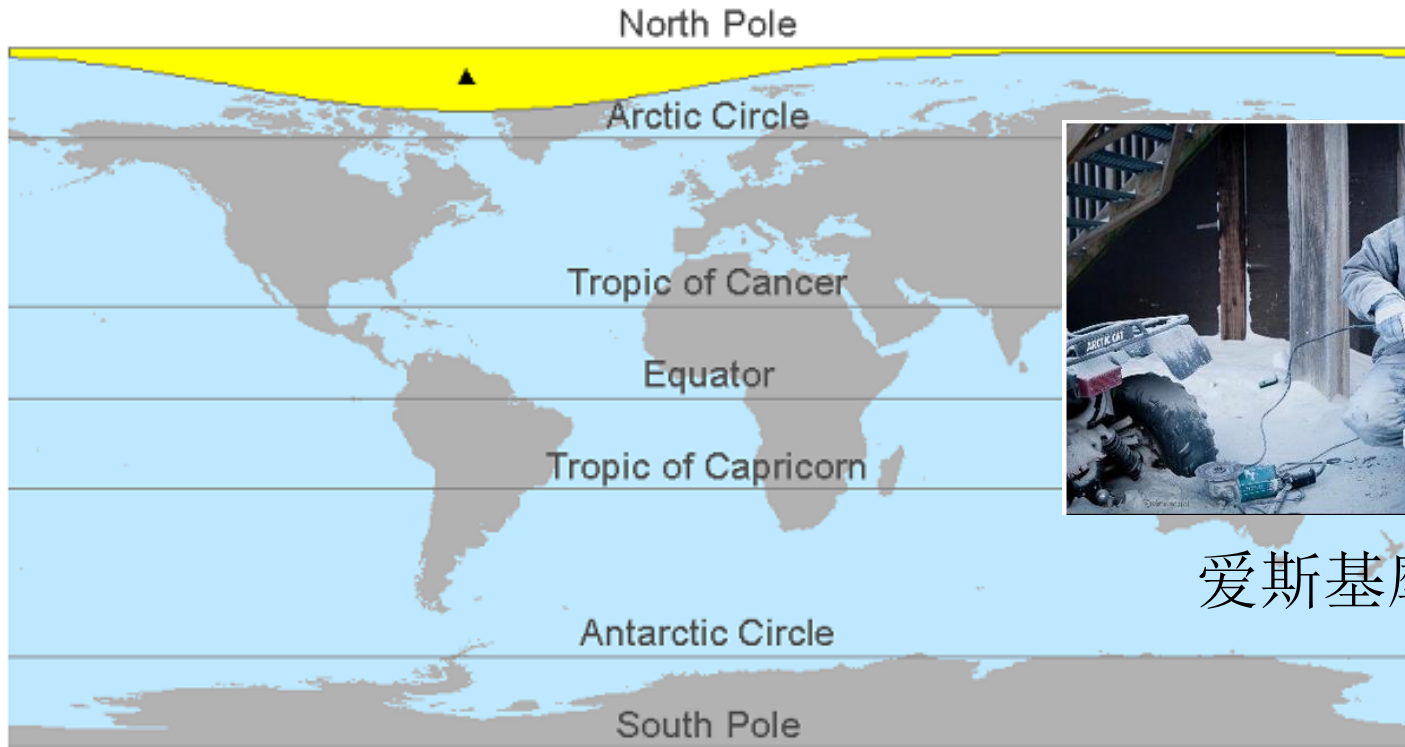


1. Buffer Analysis



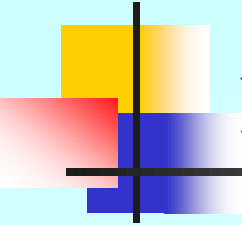
500 kilometer buffers around a selected set of world cities.

1. Buffer Analysis



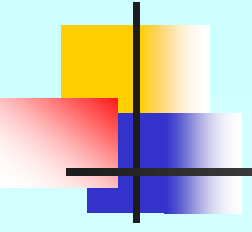
爱斯基摩人

Nunavut (努纳武特) is the northernmost permanently inhabited place in the world. Its location is represented by a black triangle below. The result of generating a "1000 kilometer" geodesic buffer around that point is represented by the yellow area.



1. Buffer Analysis

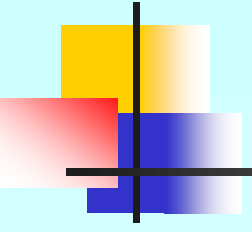
问题**3**：数据建库中的**Tolerance**参数如何影响缓冲区分析的结果？



2. Overlay Analysis

■ Basic Conceptions

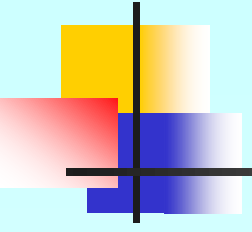
Overlay (Intersect) feature lays of several themes and so as to create a new layer, which integrated all attributes of the original features overlaid. The operation created not only new spatial relationship, but also joined all attributes and created a new attribute database.



2. Overlay Analysis

■ Tips

- The feature layers to be *overlaid* must be in the same sites with the same coordinate system.
- The overlaid layers have the same datum plane.
- The feature layers were geo-referenced exactly.



2. Overlay Analysis

■ Types of Overlay Analysis

- Erase
- Identity
- Intersect
- Symmetrical difference
- Union
- Update

2. Overlay Analysis

■ 图层擦除 (Erase)



多边形与多边形



线与多边形



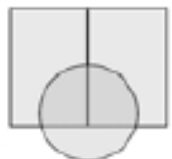
点与多边形

$A - A \cap B$ (即 $x \in A$ 且 $x \notin B$, A 为输入图层, B 为擦除层)

2. Overlay Analysis

■ 识别叠加 (Identity)

输入图层



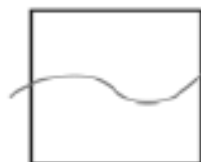
有相同特征



输出图层



多边形和多边形



输入图层



输出图层

线和多边形

输入图层



输出图层

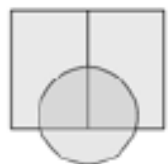


点和多边形

2. Overlay Analysis

■ 交集操作 (Intersect)

输入图层



相交的图层

输出图层



多边形与多边形

输入图层



输出图层



线与多边形

输入图层

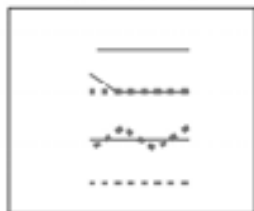


输出图层



点与多边形

输入图层



输出图层



线与线

输入图层



输出图层



点与线

输入图层



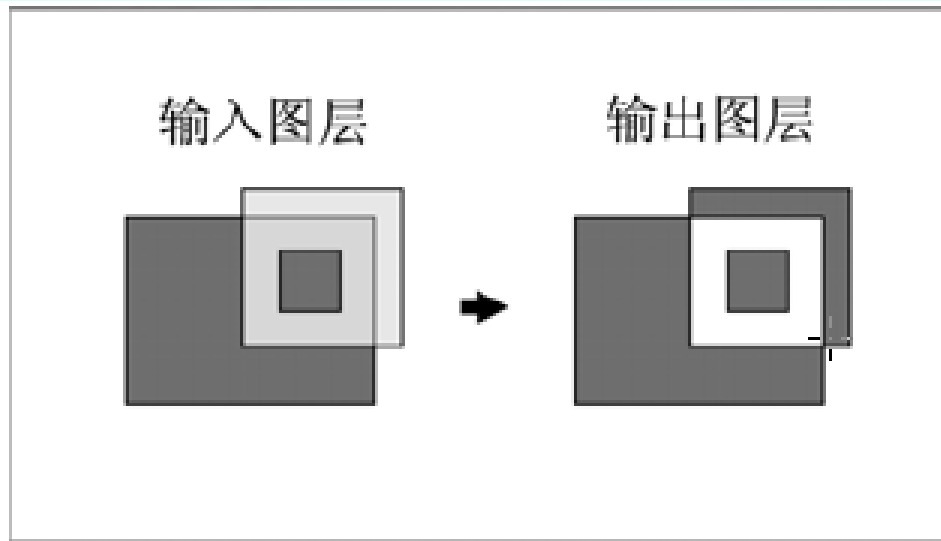
输出图层



点与点

2. Overlay Analysis

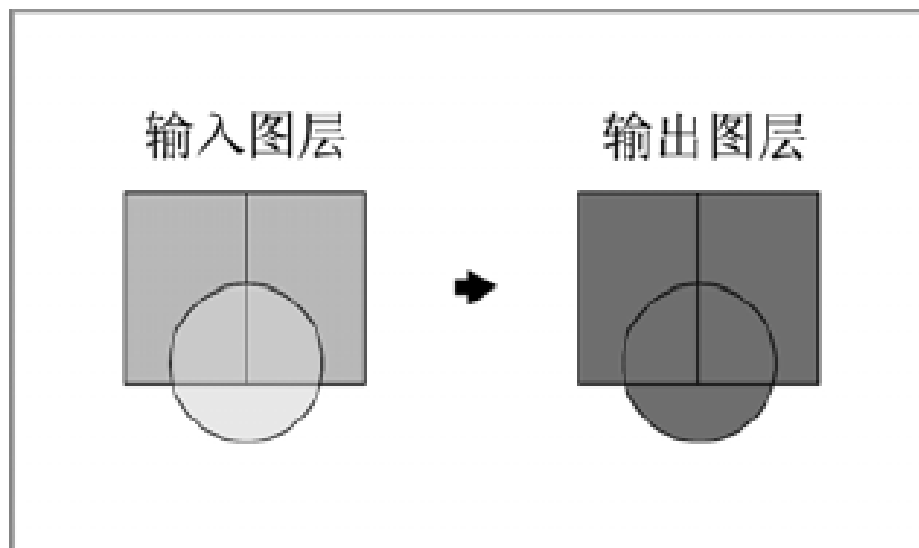
■ 均匀差值 (Symmetrical difference)



$$x \in (A \cup B - A \cap B)$$

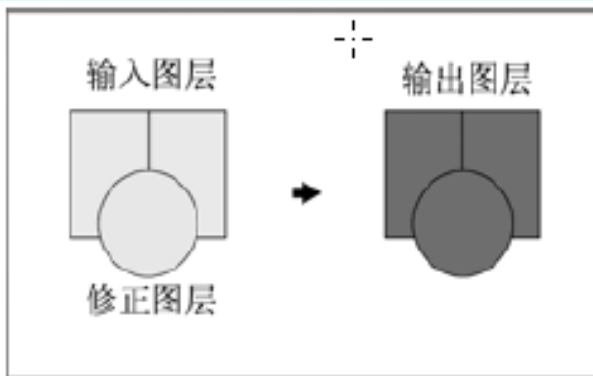
2. Overlay Analysis

■ 图层合并 (Union)

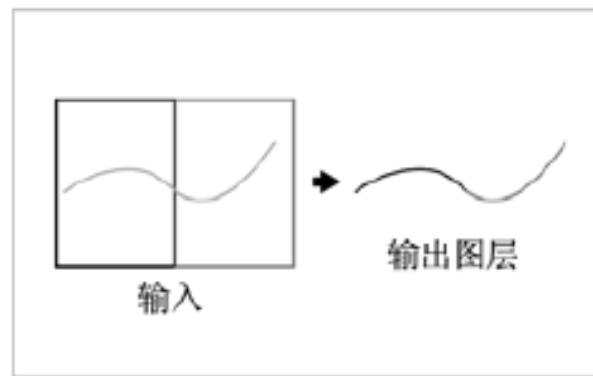


2. Overlay Analysis

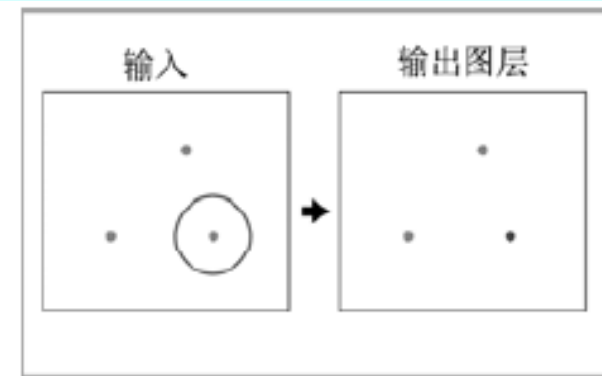
■ 修正更新 (Update)



多边形与多边形



线与多边形



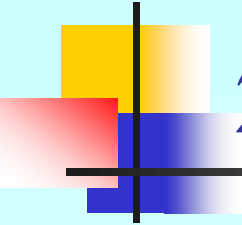
点与多边形



■ not merge

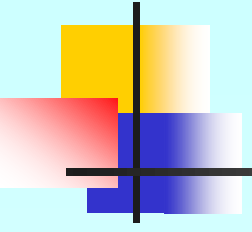
■ merge





2. Overlay Analysis

Input data Erase Features	point	polyline	polygon
point	/	/	/
polyline	/	/	/
polygon	√	√	√



2. Overlay Analysis

- Common Problems (FAQ)

- The change of attribute database

- Fragment Polygons

